

Deyang (Dylan) Hsu

U.S. Citizen | Open to Relocation | (380) 218-5944 | deyang.hsu@gmail.com
linkedin.com/in/deyang-hsu | github.com/deyangh | deyangh.github.io

Education

University of Southern California, Viterbi School of Engineering

M.S. in Computer Science | GPA: 3.88/4.0

Los Angeles, CA

2024 – 2026

The Ohio State University

B.S. in Electrical & Computer Engineering | GPA: 3.77/4.0 | *Magna Cum Laude*

Columbus, OH

2018 – 2022

Professional Experience

Intel Corporation

Software Development Engineer

Chandler, AZ

2022 – 2023

- Root-caused and resolved P0 production defects (race conditions, memory leaks) across enterprise **Python** microservices, cutting service downtime by **27%** and reducing average incident time-to-resolution.
- Refactored legacy manufacturing database schemas and rewrote hot-path queries with optimized **PL/SQL** and indexing strategies, reducing query latency and scaling write throughput for high-volume data streams.
- Designed, tested, and deployed automated **Python** wafer lot-tracking software adopted across global fabs, driving an **18% increase** in wafer throughput under strict quality-compliance standards.

General Motors

Manufacturing Controls Engineer Intern

Bay City, MI

2021

- Debugged hardware and software runtime faults in robotics systems and PLCs, cutting plant assembly downtime by **8%**.
- Integrated an IO-Link smart pressure sensor network via **Siemens SIMATIC STEP 7**, reducing inventory scrapping by **13%** and catching upstream calibration defects early.

Projects & Teaching

AgentGauntlet — Security Test Harness for LLM Agents

github.com/deyangh/agentgauntlet | [demo](#)

- Designed and shipped “pytest for agent security” (Python, Apache-2.0): a test harness that runs any LLM agent through **16 adversarial scenarios** across **8 attack categories** (prompt injection, data exfiltration, confused-deputy, tool misuse), with a **pytest plugin** that fails CI on security regressions.
- Architected a harness-owned agent loop with canary-token detectors, so attack success is an *observed tool call*, not an LLM judgment; dual-axis scoring (resistance $\times$ task completion) blocks refuse-everything gaming.

Signet — Cross-Language LLM Watermark Detection Service

github.com/deyangh/signet

- Built a **Go** HTTP detection service (bounded worker pool, Prometheus metrics, dependency-free container) against a **Python** generation library, specifying a keyed PRF (HMAC-SHA256 $\rightarrow$ ChaCha20) so both languages agree bit-for-bit under CI-enforced parity tests; **5 $\times$** throughput on 10 cores, per-request allocations cut from **2,401 to 214**.

StressID: Multimodal Physiological Stress Detection

PyTorch, Wav2Vec2, MediaPipe

- Engineered a three-branch (audio, face, gesture) late-fusion pipeline on the **NeurIPS 2023 StressID** dataset — custom CNN + Transformer encoder with learnable attention pooling — validated via subject-level stratified 5-fold cross-validation under 71% class imbalance, beating frozen Wav2Vec2 baselines.

The Ohio State University — Dept. of Electrical & Computer Engineering

Columbus, OH

Teaching Assistant / Grader, Advanced C++ Programming

2021 – 2022

- Graded **C++** projects for **60+** students; ran office hours debugging segfaults and memory leaks (GDB, Valgrind).

Research & Publications

FORTIS Lab — University of Southern California

Graduate Research Assistant (Advisor: Prof. Yue Zhao)

Los Angeles, CA

2025 – 2026

- Architected a spatio-temporal wildfire-prediction pipeline ingesting and fusing **116,000+** observations from heterogeneous geospatial sources (ERA5, LANDFIRE), achieving a **15% F1 gain** over baselines.
- Profiled and optimized deep-learning training on **NVIDIA A100s** (mixed precision, `torch.compile`) for a **20 $\times$ speedup** — 8+ hours to ~25 minutes per run — enabling rapid architecture ablation.
- Developed automated LLM evaluation pipelines benchmarking GPT-4, Claude, and DeepSeek on multi-narrative conflict detection, and **H-MAD**, an asynchronous multi-agent debate framework quantifying linguistic bias in open-weight models.

The Traffickers’ Pitch: Detecting Deceptive Recruitment in Online Job Boards

ICWSM 2026

S. Zhou, P. Qiu, T. Salkar, L.B. Urrutia, D. Shen, **D. Hsu**, E.C. Choi, E. Ferrara — [arXiv:2605.25416](https://arxiv.org/abs/2605.25416)

- Co-authored a deep learning framework using contextual embeddings to isolate fraudulent recruitment patterns; engineered scalable crawling and regex pre-processing pipelines over **50K+** job postings.

Technical Skills

Languages

Python, Go, C, C++, Java, SQL (PL/SQL), Bash

Software Eng.

Data structures & algorithms, OOP design, microservices, REST APIs, unit & integration testing (pytest), CI/CD, code review, Agile/Scrum, debugging & profiling (GDB, Valgrind)

Systems & Infra

Linux, Git, GitHub Actions, Docker, Kubernetes, Prometheus, Oracle, MySQL, concurrency & multithreading (goroutines, OpenMP, MPI), CUDA, NVIDIA A100 (AMP, `torch.compile`), SLURM

ML & Modeling

PyTorch, TensorFlow, HuggingFace, Transformers, LiteLLM, LLM evaluation, GNNs, CNNs, LoRA/PEFT, RAG, Anomaly Detection, scikit-learn, XGBoost